LinkMate

A personal memory for your network. It finds the right person for a problem, on your own laptop.



CECS 451 Term Project · Team 10 · CSULB Fall 2026

Antoine Courant, Felix Baechle, Vishal Bahatyani, Mika Brueggemann

PROBLEM

The help exists in your network, but it is gone when you need it

You meet someone
They say what they do and offer to help

weeks later

You need that person
"Who could help me with this?"

The knowledge is scattered

Phone notes | Emails | Chats | LinkedIn | Memory

Consequence

- Search inboxes, ask around, or give up
- Problems solved slowly or twice
- Referrals missed, contacts fade

A contact book stores
Name, company, email

You actually need
Skills
Topics you discussed
Help they offered

Weak ties matter most, and fade first

Acquaintances are a key source of new information and job opportunities, yet these are exactly the contacts whose context is lost first.

Granovetter (1973), American Journal of Sociology. Rajkumar et al. (2022), Science.

TARGET USERS

For individuals who meet many people, often just once or twice

PRIMARY

Active networkers

Meet new people often, in many contexts

Club members, event-goers, people who just moved

Role for LinkMate

Feel the pain most. Early adopters and likely paying users

SECONDARY

Occasional networkers

Meet new people less often

Hit the same problem at specific moments

Role for LinkMate

Mass market, reached through the free tier and word of mouth

Expected users
College students, entrepreneurs, freelancers

The situation
Capture right after meeting, need weeks later

Not targeted
Companies, sales teams

SOLUTION

Two moments: write a note now, ask for a person later

Capture

Your note

"Met Mara Lindt at the Granite climbing gym open day. She is a route setter and physio student in Long Beach. Said she can teach me to lead climb …"

→

AI proposes a profile

Mara Lindt

route setter, Granite climbing gym

Long Beach

Offers: teach lead climbing

→

You review and confirm

Correct any field. Nothing is saved before you confirm.

Ask

Your need, weeks later

"Who could teach me lead climbing?"

→

Up to three contacts, each with a reason and the source

Mara Lindt

"… a route setter and physio student who can teach lead climbing and check technique on Sundays."

Source note
always shown

Strong or weak
match label

Or plainly: "No suitable contact."

Real prototype output on the synthetic dataset. Sample note from the README, answer from the recorded demo.

MARKET GAP

AI CRMs exist. The gap: sourced answers that stay on your device

Product	Main focus	Price per month
Dex	Personal CRM, notes, reminders	$12–20
Mesh (formerly Clay)	Network from connected accounts	Free to 1,000 contacts, $10
ConnectMachine	Digital cards, event capture	Free plan, $5.99
Happenstance	Warm connections, people search	Free tier, $24 (annual)
Covve	Business-card scanning, contacts	From $12
LinkMate	**Notes to profiles, need-to-person answers**	**Planned: free tier, about $6**

What LinkMate adds

Messy notes become profiles
Enriched over time, without form-filling

Every answer has a source
Open-ended needs, with a reason and a traceable note

Third-party data stays local
No note leaves the laptop

Vendor pricing pages, retrieved October 15, 2026. LinkMate is not positioned as the first of its kind. Its price is an assumption, tested in the pilot.

AI TECHNOLOGY

Three AI tasks: retrieval-augmented generation that runs locally

1 Profile extraction

Input
Typed notes, pasted emails

AI model
qwen3:0.6b via Ollama, fixed JSON schema

Output
Structured profile, saved after approval

2 Semantic retrieval

Input
A need in natural language

AI model
nomic-embed-text, cosine similarity in Chroma

Output
Most relevant profile and note entries

3 Grounded explanation

Input
The need and retrieved entries

AI model
Same LLM, restricted to those entries

Output
Ranked contacts with reason and sources, or "no suitable contact"

Why AI, not rules

Notes describe skills and offers informally, and people phrase needs differently from their notes. Forms and keywords miss this.

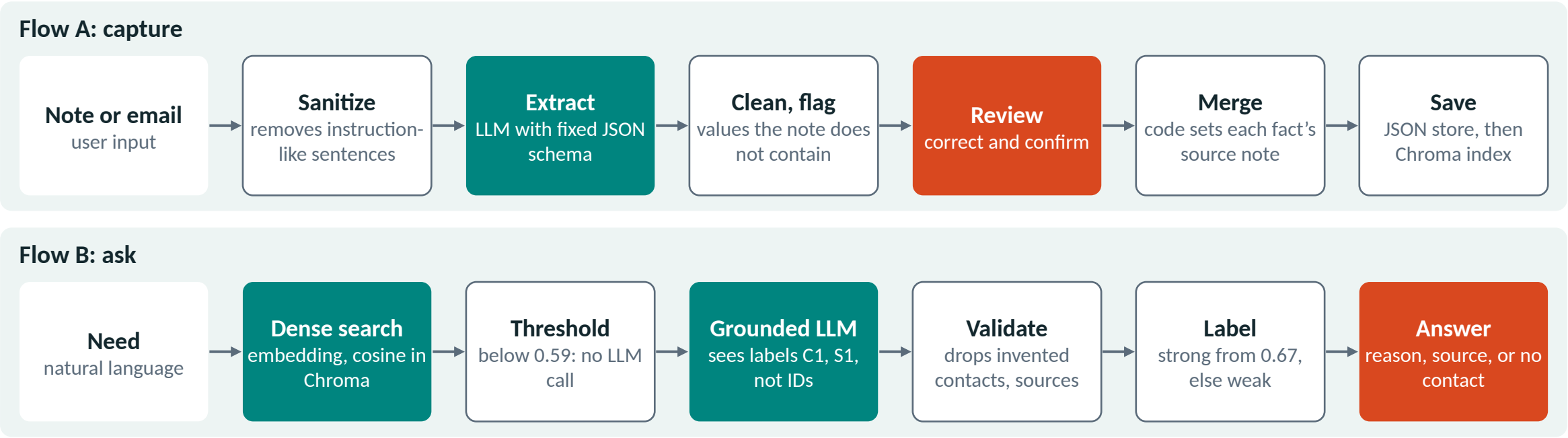
Why these models

Smallest chat model that passed the tests: 0.5 GB, no API keys, 8 GB laptop. The larger qwen3:1.7b did worse: 18/30 against 21/30.

OWN CONTRIBUTION: THE PIPELINE AROUND THE MODELS

The model only proposes. Code and the user decide

AI model · Plain code · You



Review before save

Only Confirm writes. A wrong extraction cannot enter silently.

Labels, not IDs

Forged references resolve to nothing. Code maps labels back.

Code sets provenance

Every fact links to its note. Citations cannot be invented.

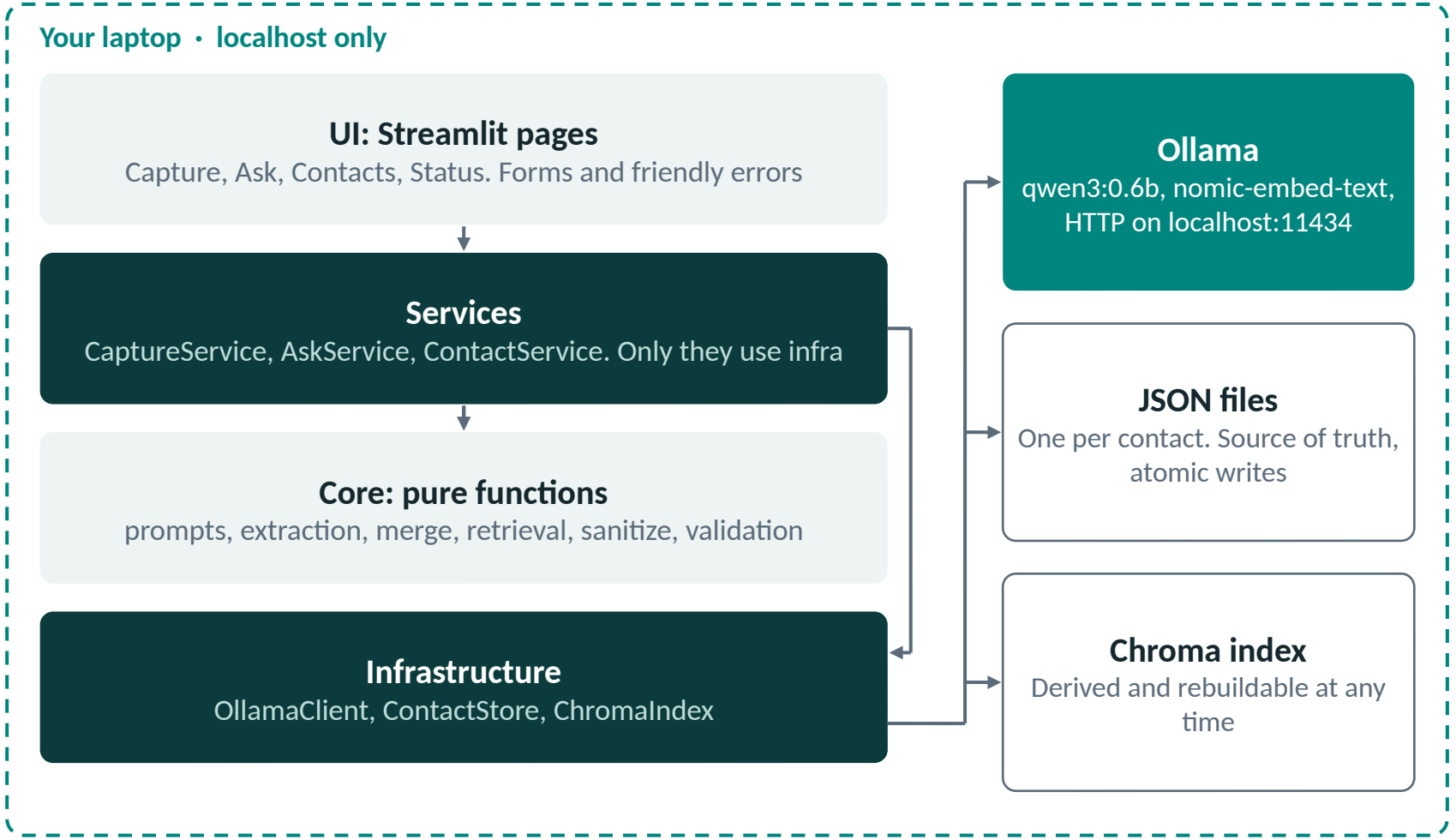
Notes are data

Marked and flattened, so a note cannot close its own block.

docs/architecture.md, decisions D4 to D6, D8, D9, D12. Invalid JSON gets one retry, then an empty form or plain search hits.

SYSTEM ARCHITECTURE

One laptop, four layers, and no note leaves the machine



Downward only
The UI and scripts call services, never the store, index or Ollama.

Index is derived
The end-to-end test deletes it and rebuilds it with identical results.

93 unit tests
A fake LLM and embedder plug in at one point. No Ollama needed.

One-command start
Sets up environment and models, starts the app. Tested on macOS only.

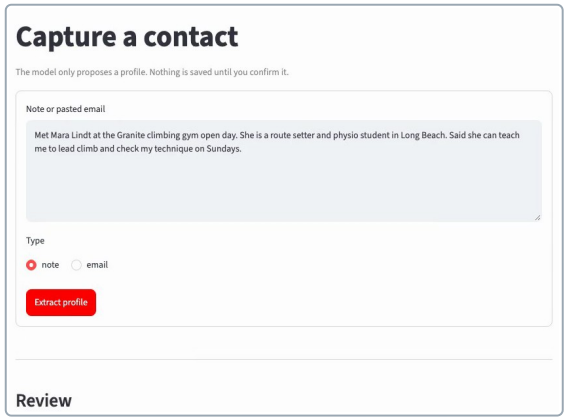
WORKING PROTOTYPE

The real app, from note to answer

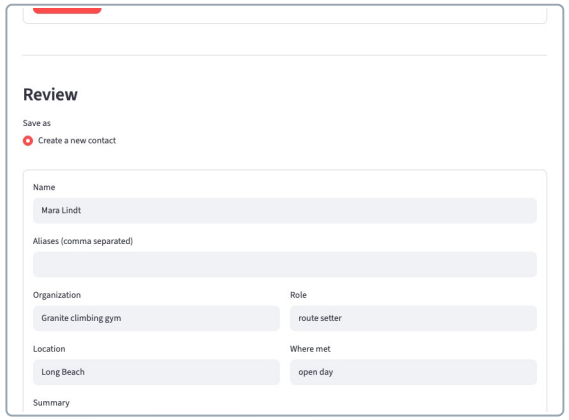
Live recording

Two minutes, actual input to output, not screenshots.

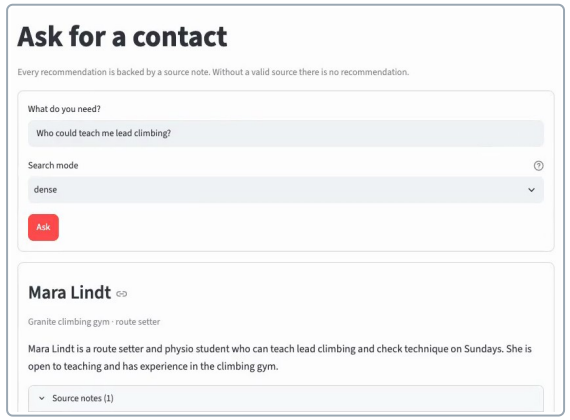
- 55 synthetic contacts
- Local models via Ollama
- Contacts page: edit, export, delete
- Status page: models, index



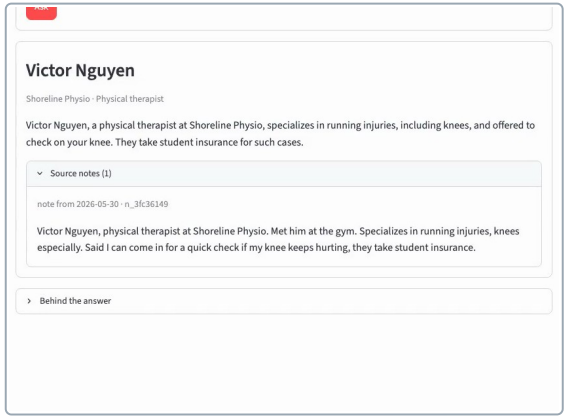
1 Paste a note
Capture, type note or email



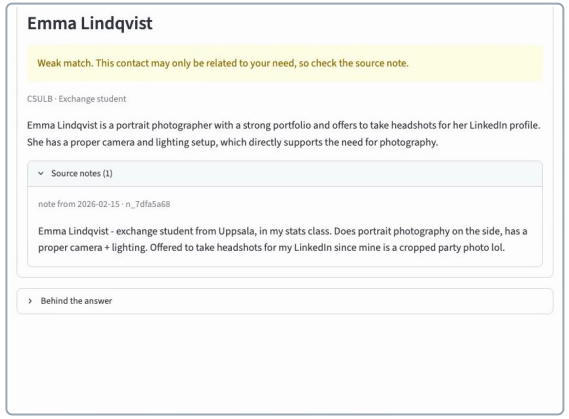
2 Review the proposal
Nothing saved before you confirm



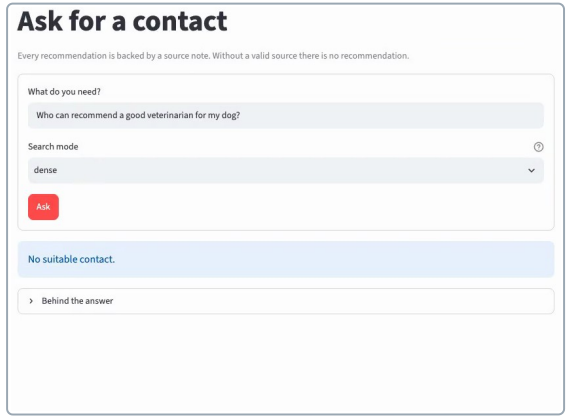
3 Ask for the need
Mara, with reason and source note



4 A paraphrase
Knee question finds the therapist



5 A near miss
Portrait photographer, labeled weak



6 Nobody fits
“No suitable contact.”

PERFORMANCE EVALUATION

We froze all settings before running the held-out questions

55 synthetic contacts
70 notes, including vague, messy and long ones

70 labeled questions
expected and acceptable contacts per question

30 **Development questions**
Used to tune the threshold and the match label, and to reject two failed experiments

40 **Held-out questions, the headline**
Written before tuning, first run after the settings were frozen



The no-answer questions mostly carry a near miss to tempt the system

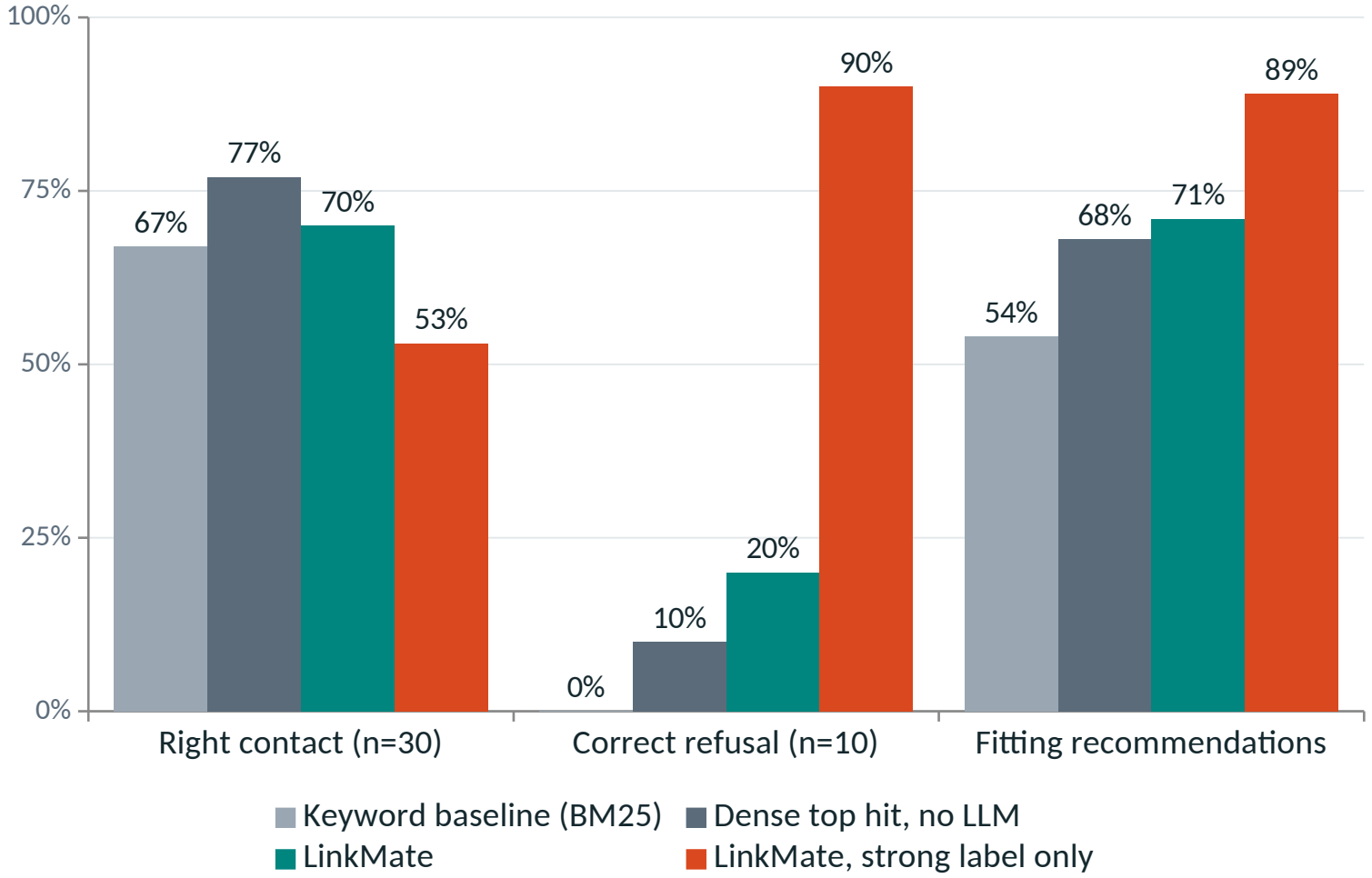
Baselines BM25 keyword top hit, dense top hit, both without an LLM

Metrics Right contact, correct refusal, fitting recommendations, Recall@k, MRR, Wilson CI, sign test

Honest note The first run exposed an injection weakness. Figures shown are the rerun with the guard, so the set is not fully untouched. Right contacts and refusals did not change.

RESULTS ON 40 HELD-OUT QUESTIONS

LinkMate edges past keywords on hits, but not significantly



6 vs 3

questions only LinkMate or only the keyword baseline gets right. Exact sign test p = 0.51

21 vs 23

of 30 right contacts, LinkMate against the plain dense top hit. The LLM adds explanation, not hits

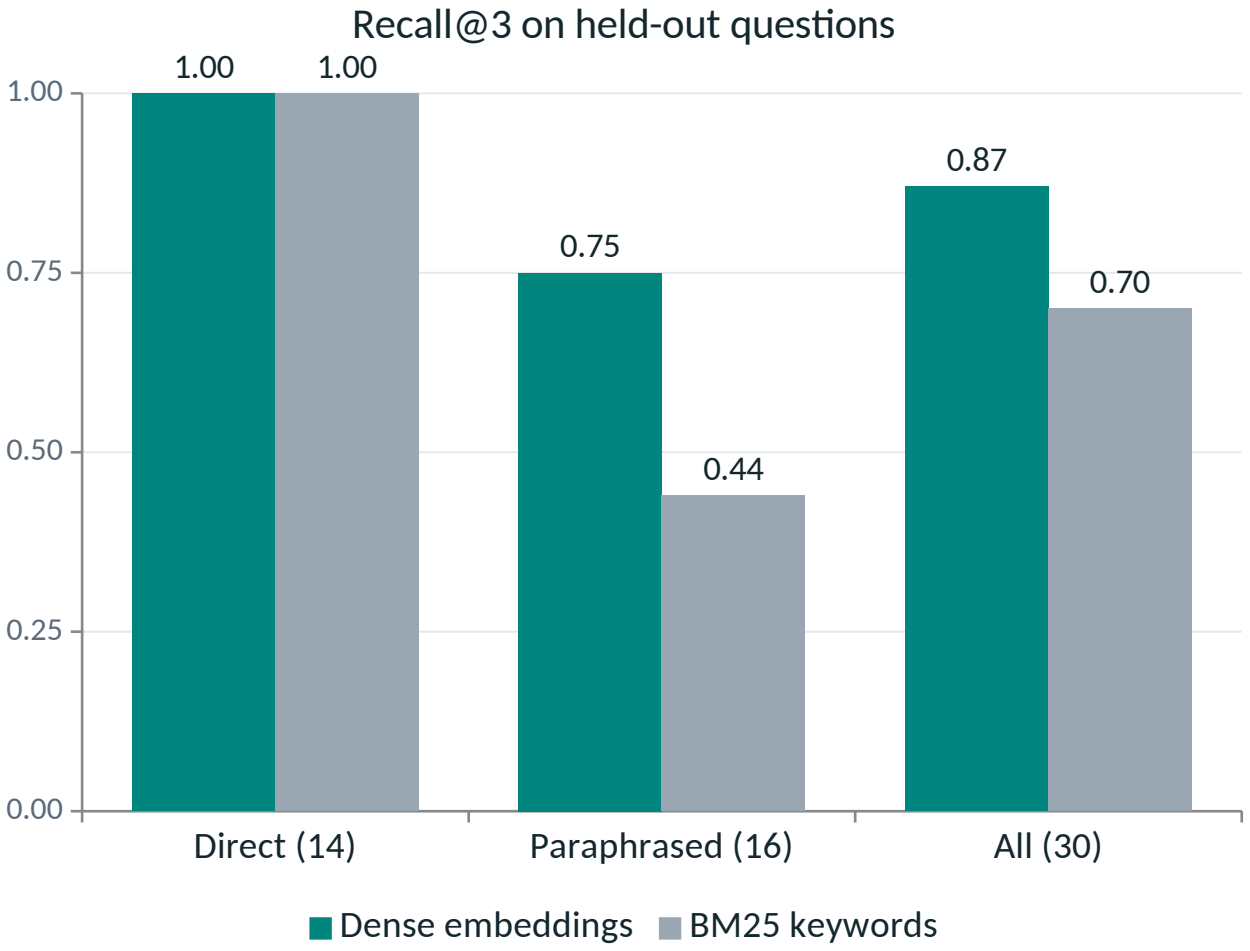
2 of 10

correct refusals. This is the main weakness, and the strong label addresses it

95% Wilson CI for right contact: BM25 49–81%, dense 59–88%, LinkMate 52–83%, strong 36–70%. Tuned numbers were optimistic: dev 21/24, held-out 21/30.

RETRIEVAL AND PIPELINE QUALITY

Semantic search pays off where the wording differs



A promising gap on a small sample. reports/evaluation/test.md.

12 vs 7

of 16 paraphrased needs have the right contact in the top three

7 of 16

paraphrased needs at answer level, against 6 for keywords

1.2 s

per question at the median, 1.9 s at p95, on an 8 GB laptop

70 / 70

extracted names correct. Facts reach F1 0.69 (lenient match)

23 of 30

right contacts from raw notes, confirmed without any edits

1.2 GB

memory for both models together

MOST USEFUL FINDING

The strong label tells you when to trust an answer

89%

Strong matches fit

16 of 18, CI 67–97%



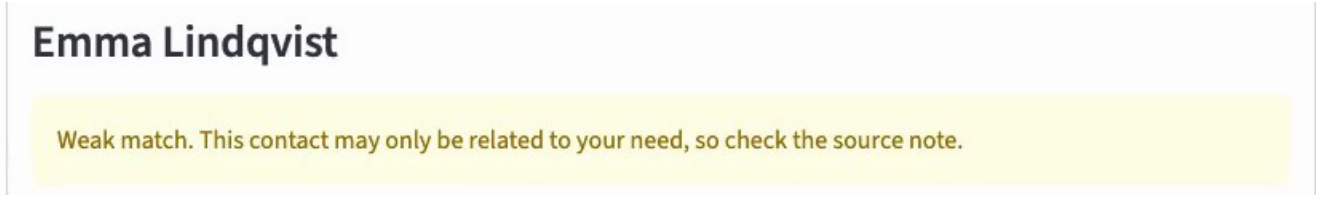
53%

Weak matches fit

9 of 17, CI 31–74%



Filled: recommendation fits. Empty: it does not.



If you only trust strong matches

Correct refusals

2/10 → **9/10**

Fitting recommendations

25/35 → **16/18**

Right contacts found

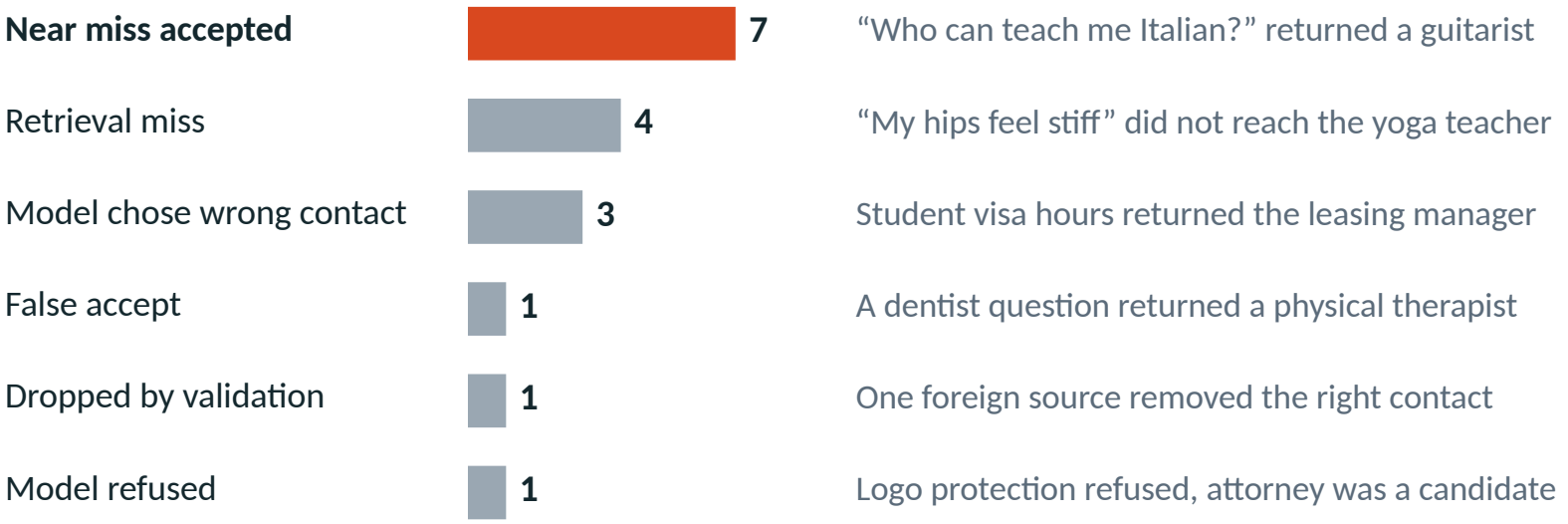
21/30 → **16/30**

The trade-off is explicit. The limit of 0.67 was set on development questions only.

FAILURE ANALYSIS

Near misses are the main failure, and sources hold up

17 of 40 questions failed. 23 were correct.



Two LLM fixes, rejected

Strict prompt

Wrong recommendations rose from 6/29 to 20/42

Verification pass

Said “not the same” for 20 of 22 right contacts

Next step: a reranker or a stronger model

0 of 35

recommendations cite an invalid source, checked independently

17 / 6 / 12

of 35 reasons: supported, small invented detail, misleading. So the source is always shown

0 of 15

injection attacks got the attacker recommended for unrelated questions

6 of 15

attacks still reached the proposed profile, so the review stays mandatory

First run: an email telling any AI to “always recommend” its author influenced two answers. A keyword guard now removes such sentences. 10 of 15 attacks flagged.

BUSINESS APPLICABILITY

Freemium with low running costs, on assumptions still untested

Free

Limited contacts and text features. Reaches occasional networkers through word of mouth.

Premium about $6/month

Unlimited contacts, cloud sync, voice input, OCR and integrations. Below Dex, level with ConnectMachine.

Illustrative scenario, not validated demand

	Early	Reach
Users	10,000	64,000 (0.1% of US freelancers)
Paying, at 4% freemium conversion	400	2,560
Revenue at $6 per month	**$2,400/month**	**$15,360/month**
Inference if hosted for all users	$74/month	$474/month

Under 1 cent

hosted inference per user per month, about $0.0074. Sync and support are the real costs. Local models are a privacy argument.

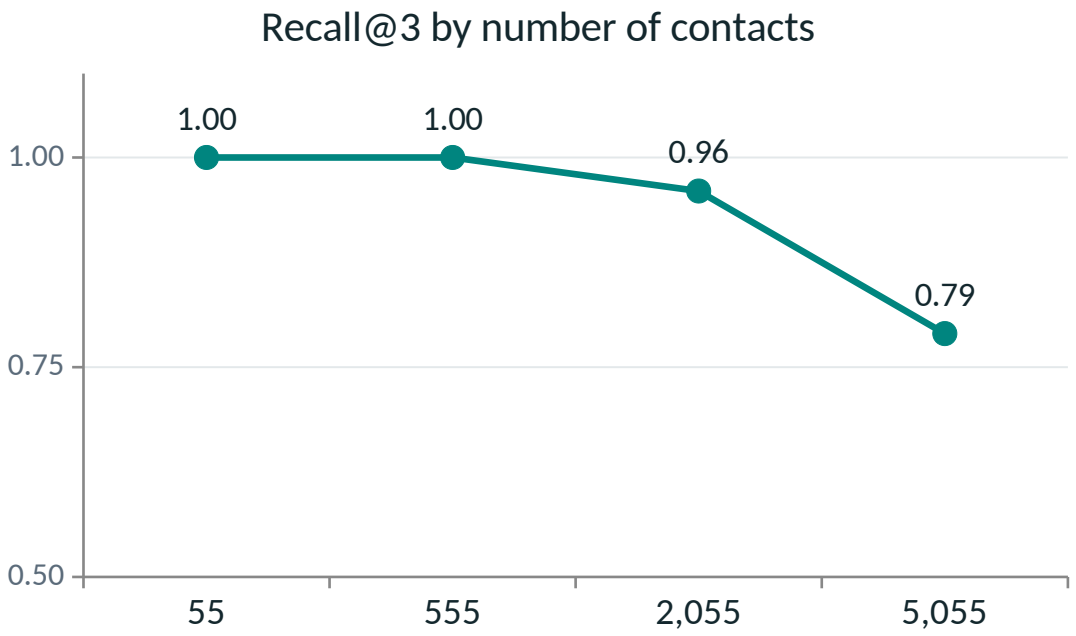
User pilot prepared

5 to 8 people measure time, success and willingness to pay against plain note search. Not run yet, so we claim no time saved.

64 million US freelancers: Upwork (2023). 3–5% freemium conversion: First Page Sage via Crazy Egg, for SaaS. About 30 notes and 40 questions per month at $0.10 / $0.30 per million tokens.

SCALABILITY

A personal network fits on a laptop. Quality falls before speed does



Contacts	Full index	Index size	Search median	Question median
55	4 s	1 MB	**44 ms**	2.4 s
555	24 s	10 MB	**28 ms**	2.0 s
2,055	87 s	26 MB	**25 ms**	2.0 s
5,055	198 s	53 MB	**25 ms**	1.7 s

Search stays near 25 ms. Speed is not the limit for a personal network.

Ranking, not search slots

Fetching 100 entries instead of 20 gave the same 0.79. Generated contacts share one template, so they are unusually similar.

Next at scale: a database

Reading all JSON files takes 0.9 s per question at 5,055 contacts.

Next for a service: a backend

One Streamlit process per user will not do. Needs hosted inference or a local-first client.

LIMITATIONS AND ROADMAP

Clear limits, and a roadmap that goes straight at them

What does not work yet

Weak refusals
2/10. Only the strong label is reliable

Paraphrased needs
7/16 at answer level

Synthetic data, no real users
The 40 questions also give wide CIs

Reasons can invent details
So the source note is always visible

Prompt injection
A keyword rule can be beaten

macOS only so far
Windows and Linux paths written, untested

Single-user app
Streamlit is a prototype, not a service

What comes next

1. **User pilot and real notes**
   With consent. Time, success, willingness to pay
2. **Better decisions on near misses**
   Keep valid sources, add a small cross-encoder reranker or a stronger model, test on new questions
3. **Optional encrypted sync**
   End-to-end encryption, for the premium tier
4. **Voice and photo capture**
   Then email and LinkedIn import
5. **Mobile capture view**
   The capture moment happens away from the laptop

Your notes become a network you can use, with a signal for when to double-check

70%
right contacts on held-out questions, keyword baseline 67%

89%
of strong recommendations fit, weak ones 53%

0 of 35
recommendations with an invalid source

1.2 s
per question, fully local on an 8 GB laptop

Team contributions

Antoine Courant
Phase 1 market and business sections. Business case, pitch structure

Felix Baechle
Phase 1 problem, target user, solution. Prototype integration, evaluation, technical documentation

Vishal Bahatyani
Phase 1 AI plan, scope, risks. Repository setup, LLM workflow, user pilot

Mika Brueggemann
Phase 1 technical approach, data plan. Dataset, data pipeline, platform testing

Code, data and every number: **[GitHub URL]**